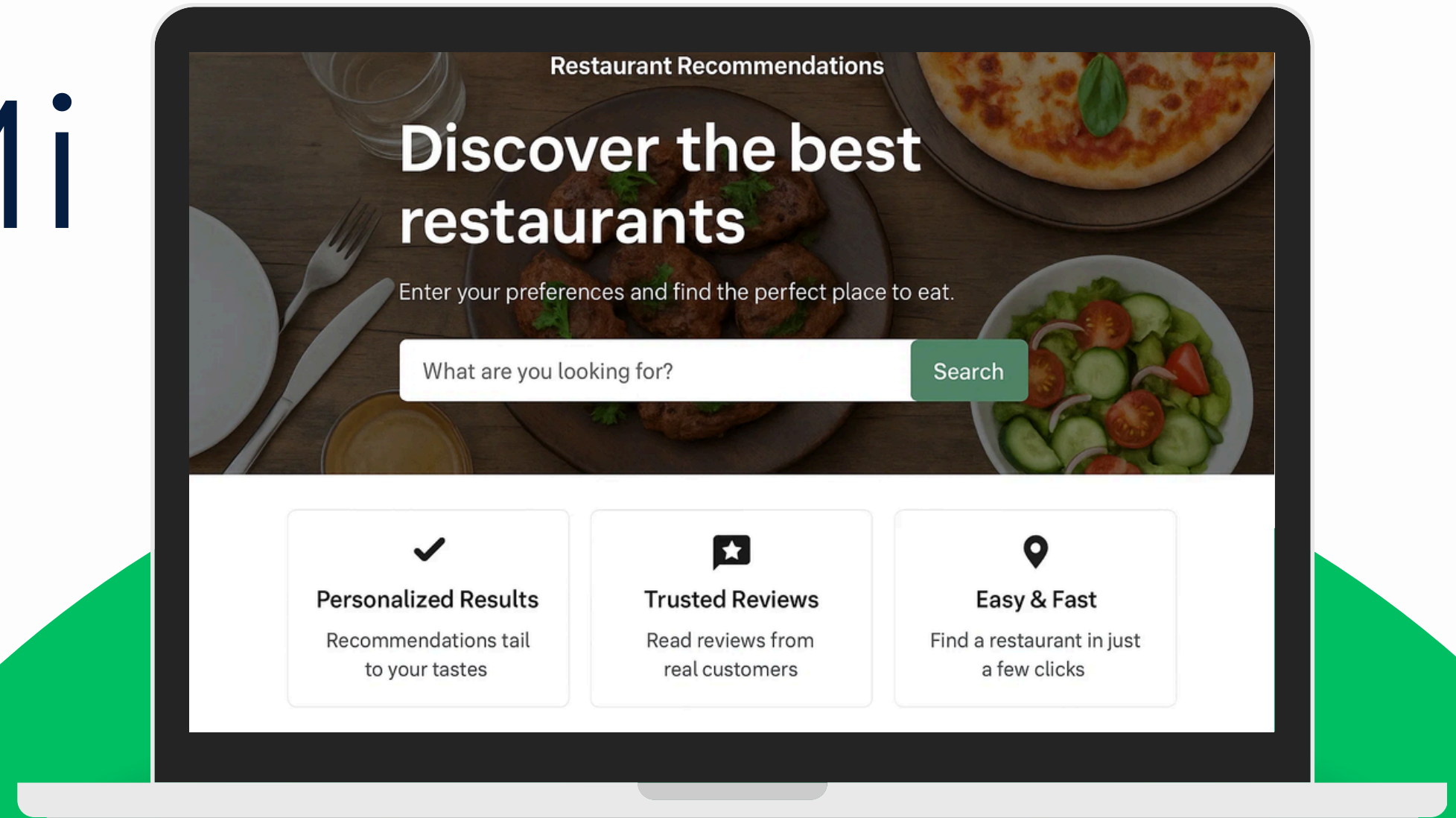


# ConsiglIA\_Mi

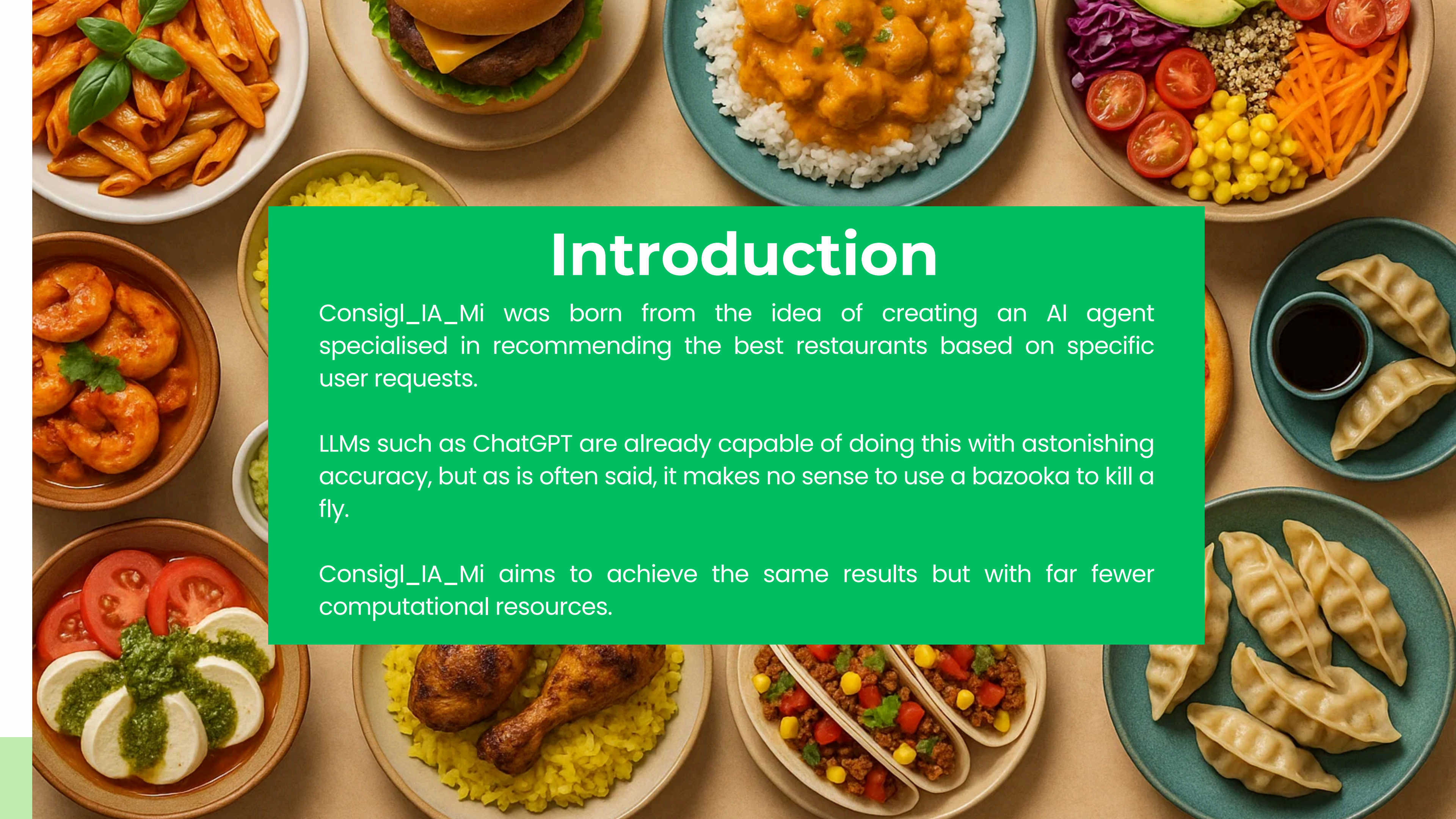
By: Jacopo Signo'



# Overview

|     |              |    |
|-----|--------------|----|
| ▶▶▶ | Introduction | 02 |
| ▶▶▶ | Goal         | 03 |
| ▶▶▶ | Methodology  | 03 |
| ▶▶▶ | Challenges   | 08 |
| ▶▶▶ | Solutions    | 05 |
| ▶▶▶ | Conclusions  | 06 |





# Introduction

Consigl\_IA\_Mi was born from the idea of creating an AI agent specialised in recommending the best restaurants based on specific user requests.

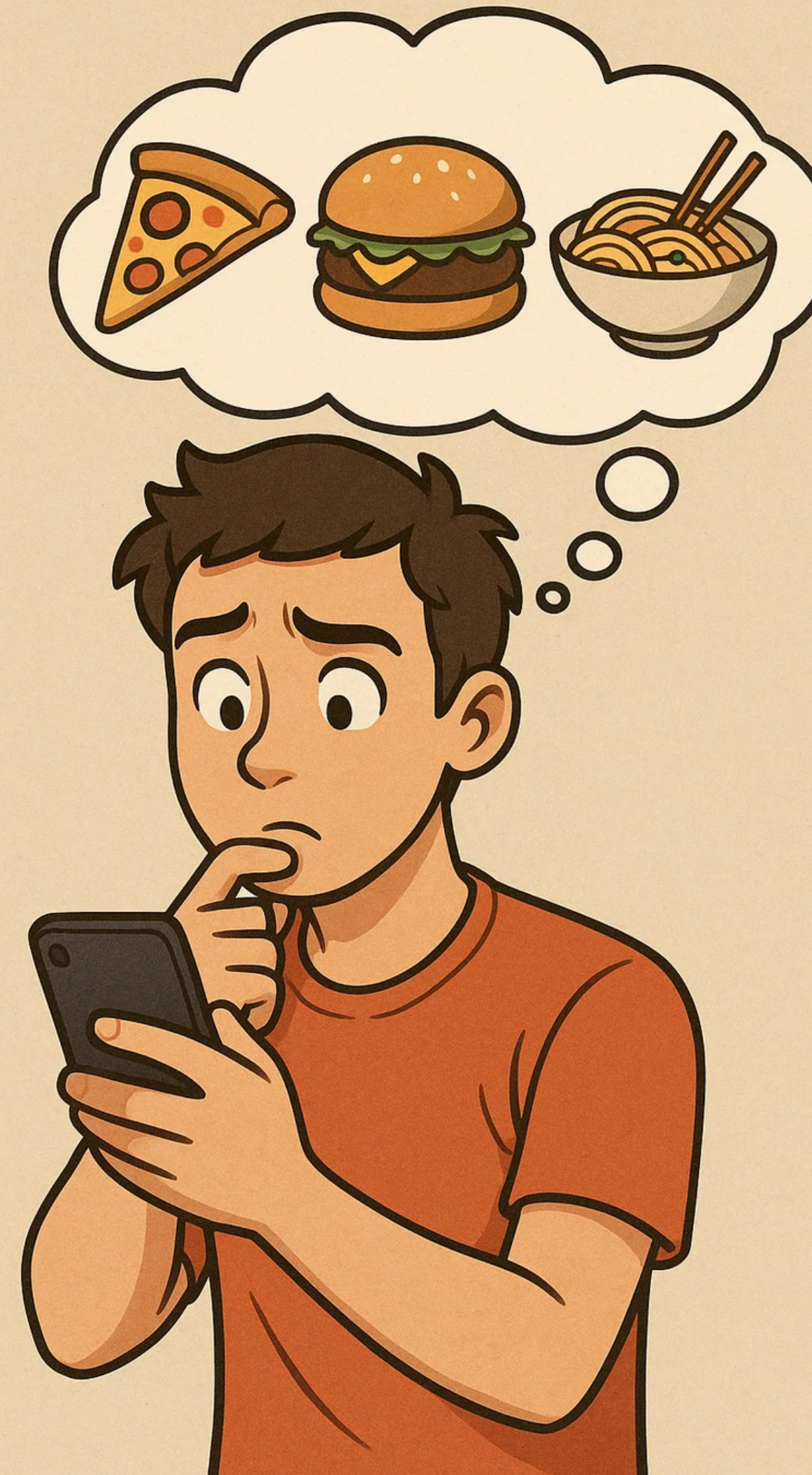
LLMs such as ChatGPT are already capable of doing this with astonishing accuracy, but as is often said, it makes no sense to use a bazooka to kill a fly.

Consigl\_IA\_Mi aims to achieve the same results but with far fewer computational resources.

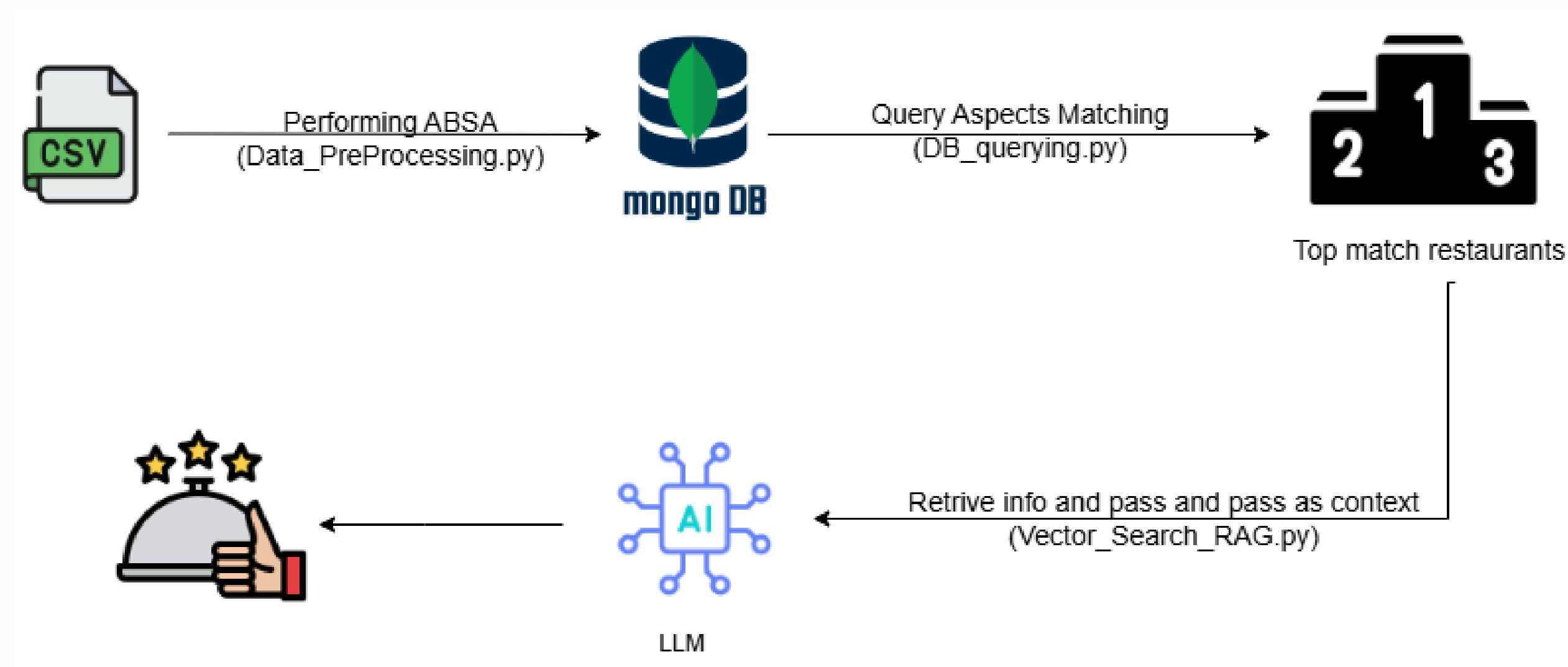
# Goal

The project was developed to create a smart restaurant recommender agent capable of tailoring its suggestions to the specific preferences and expectations of users.

*"I'm planning a dinner in Barcelona and I'm really in the mood for some good seafood – maybe **grilled octopus** or **paella**. I'd love a place where the **service is friendly**, the **atmosphere is cozy** but not too loud, and **prices are reasonable**. Also, my brother is **vegetarian**, so I'd appreciate some vegetarian tapas options as well. It'd be great if the **waiting time isn't too long**, and if the desserts, especially flan or **churros**, are worth trying, that's a big plus. Oh, and I'm sensitive to cleanliness – so that's non-negotiable. Can you help me find the best fit?"*



# Methodology



# MongoDB

```
_id: ObjectId('68a32681150e99c4db0b612b')
Unnamed: 0 : 2
parse_count : 3
restaurant_name : "Chalito_Rambla"
rating_review : 5
sample : "Positive"
review_id : "review_749758638"
title_review : "Family bonding"
review_preview : "The food is excellent.....the ambiance is very nice..the price is very..."
review_full : "The food is excellent.....the ambiance is very nice..the price is very..."
date : "March 7, 2020"
city : "Barcelona_Catalonia"
url_restaurant : "https://www.tripadvisor.com/Restaurant_Review-g187497-d13197190-Review..."
author_id : "UID_2"
▼ aspects : Object
  food : "positive"
  ambiance : "positive"
  price : "positive"
  crew : "positive"
▼ aspect_keys : Array (4)
  0: "food"
  1: "ambiance"
  2: "price"
  3: "crew"
```

# RAG template

```
prompt_template = PromptTemplate.from_template("""
    You are a food expert helping users find the best restaurants based on detailed customer reviews.

    Use ONLY the following restaurant reviews as context for your answer.

    Recommend {first_option} as the top choice. Clearly explain why this restaurant stands out using the reviews provided.

    Then, suggest these other valid alternatives: {alternatives}.
    Briefly compare them to the first option, highlighting what is better or worse based on the context.

    If the reviews are insufficient to answer, say: "I don't have enough information."

    Context:
    {context}

    User query: {question}

    Answer:
    """)
```

# Challenges

During the development process, numerous questions arose about the best approach to take.

These questions often had many possible answers, and figuring out which one was best was a very difficult process.

## ABSA Labeling

- Which model should we use?
- How to store the results?

## Reccomandetions

- How to figure out which restaurants to recommend to users?

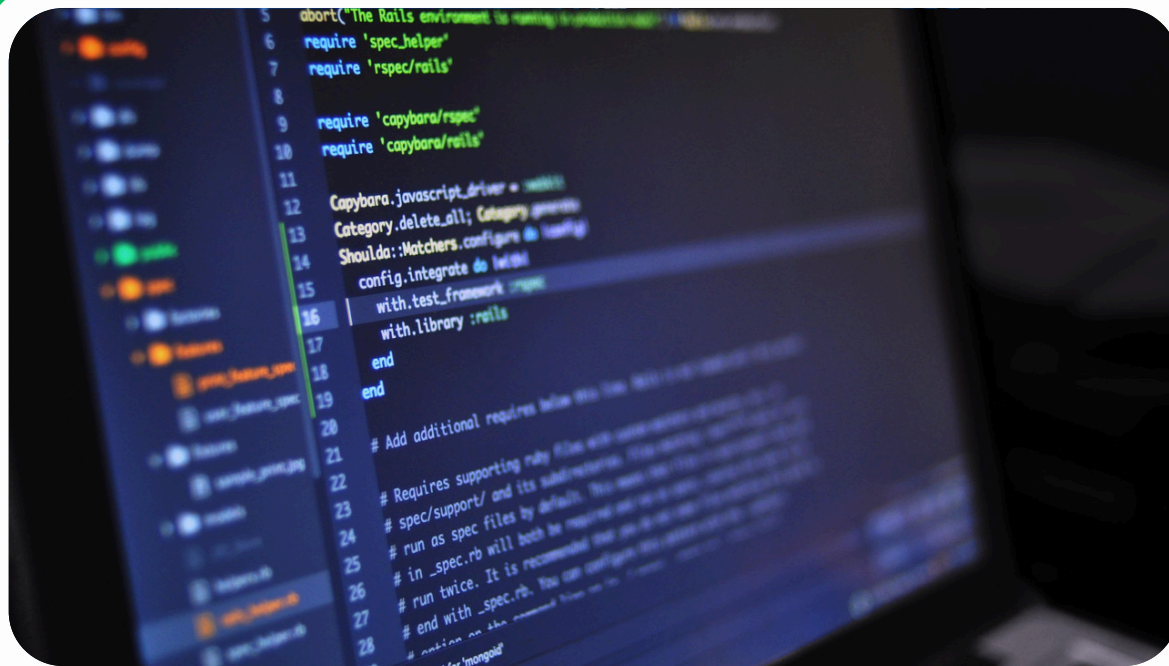
## Retrivial

- Which reviews/information to include as context in the LLM query?

## Data Storage

- How to store data efficiently?

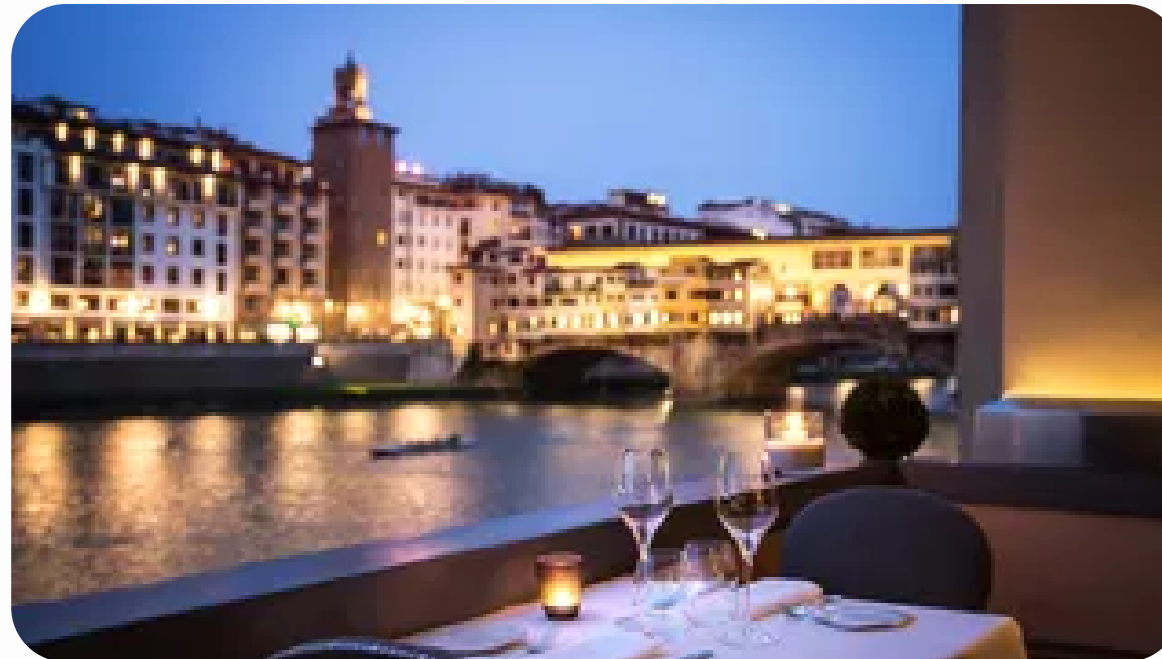
# Solutions



## ABSA Labeling

The final decision was to use a pre-trained model.

The used model is an instruction-tuned version of the T5 encoder-decoder transformer.



## Reccomandations

The best restaurants are identified using a ranking method based on how many times a restaurant appears in the review results that match the user's request.



## Retrivial

As the context passed to the LLM to formulate the final recommendation, it was decided to pass **all** reviews of the top 4 restaurants found by the 'recommendation system'.

# Conclusion

The system works and provides recommendations that are consistent with what the user asks for and that are effectively reflected in what is written in the reviews.

However, it is difficult to assess whether the advice given by the agent is actually the most desirable.

It is clear that the quality of the final output depends largely on the ability of the model responsible for performing ABSA, as the entire retrieval process starts with identifying reviews that contain certain aspects requested by the user.

# THANK YOU!

Jacopo Signo'